

Renewable Energy Auctions

Progress in Sub-Saharan Africa

Wikus Kruger

University of Cape Town

Africa Energy Forum

20 June 2018



@kruger_wikus

www.gsb.uct.a.za/mir

Three main ways to procure IPPs

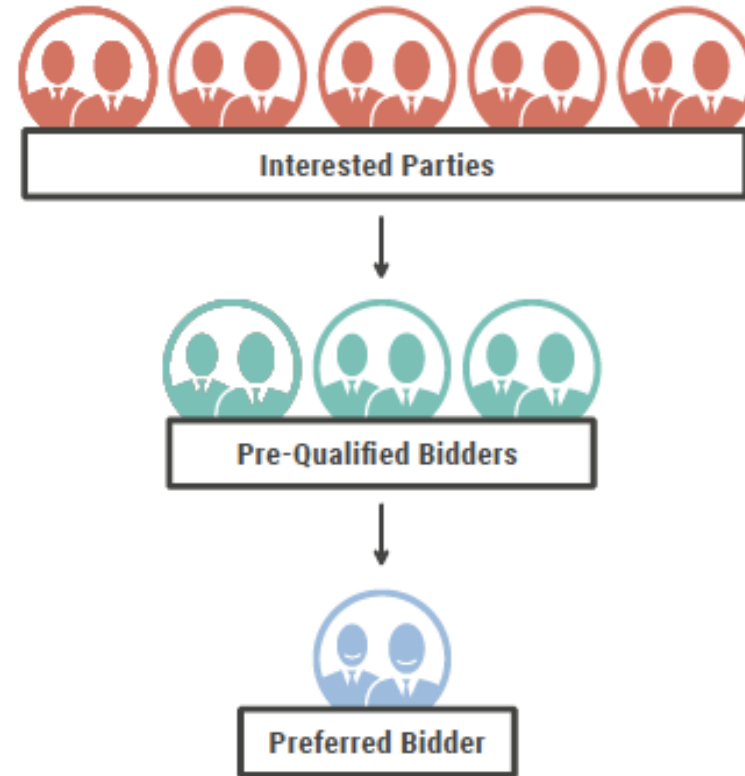
Direct Negotiations (DN)



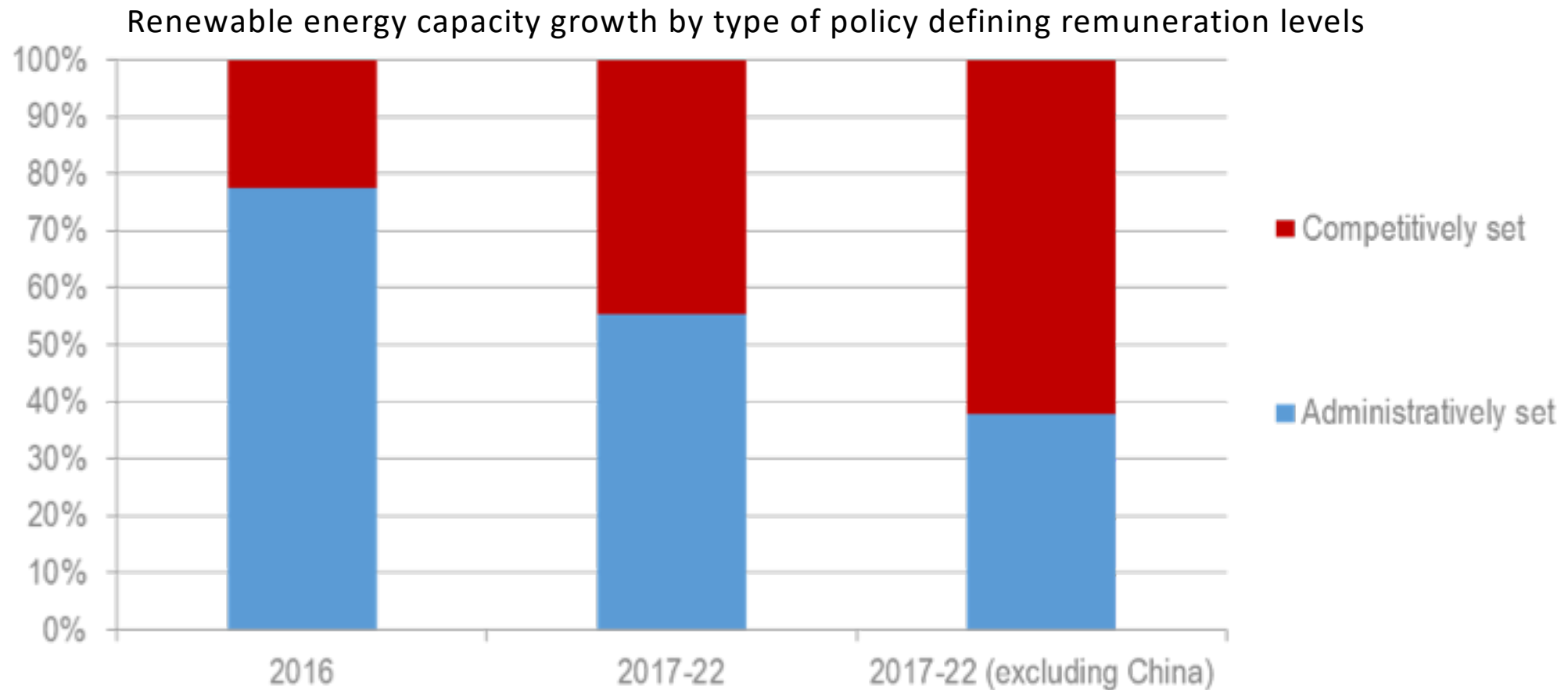
Feed-in Tariffs (REFIT)



Competitive Tenders/Auctions (ICB)

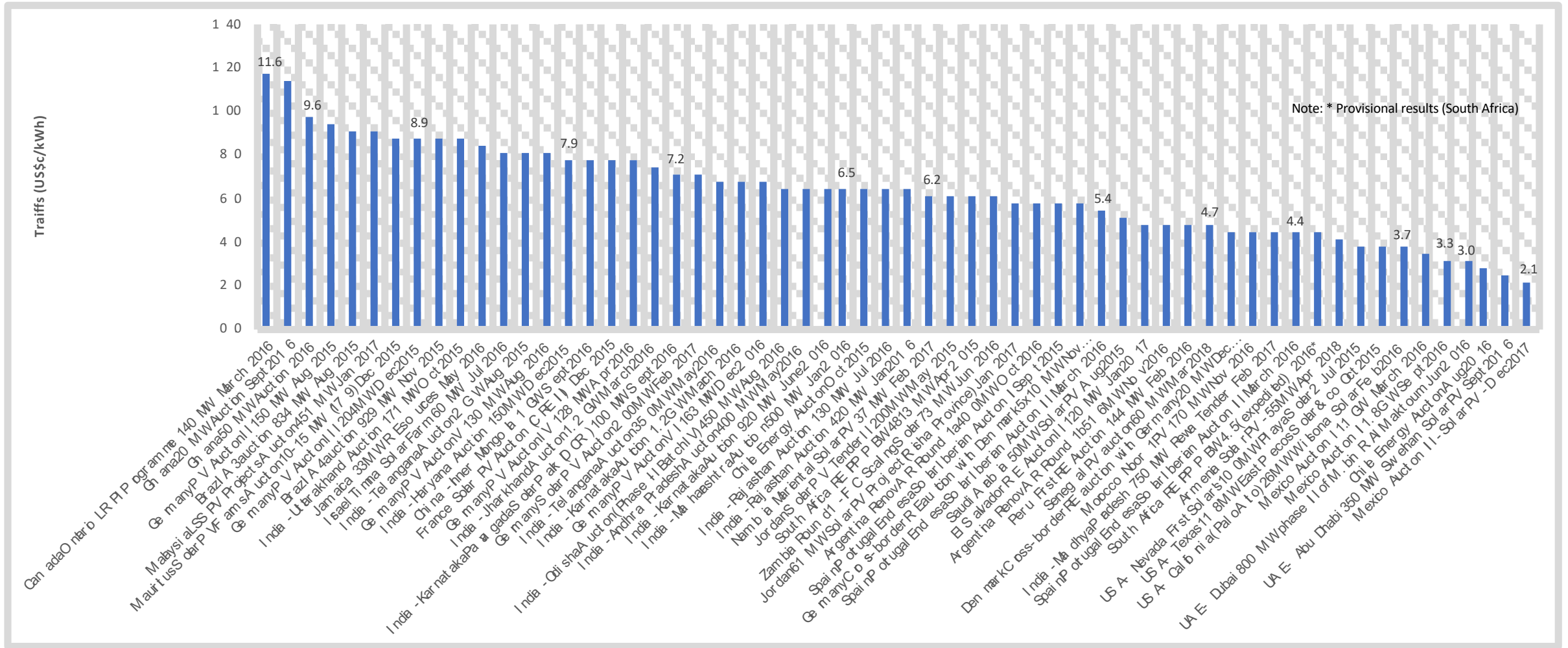


Majority of global new renewable energy capacity to be procured through auctions

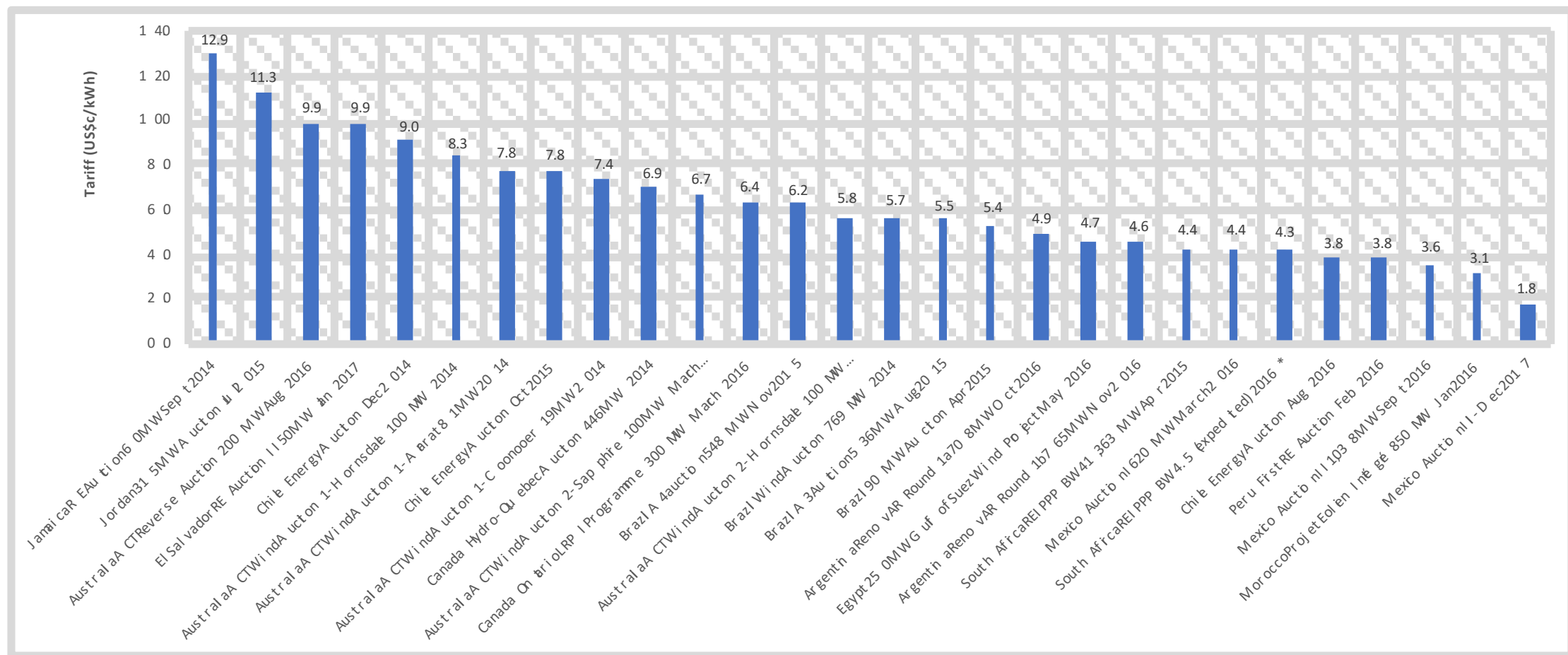


Source: IEA, 2018

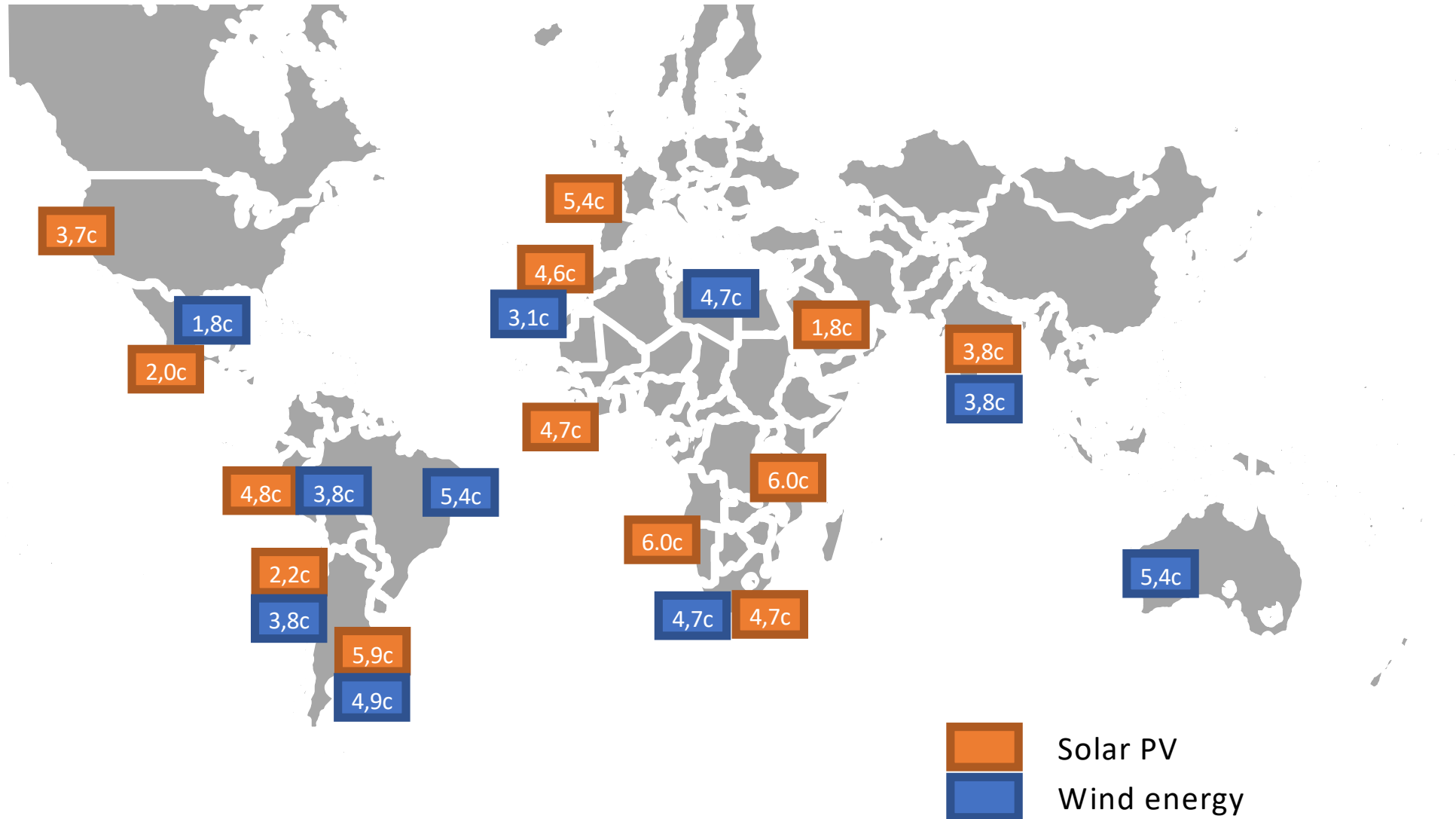
Auctions are exerting significant downward price pressure on solar PV projects



...and on wind projects



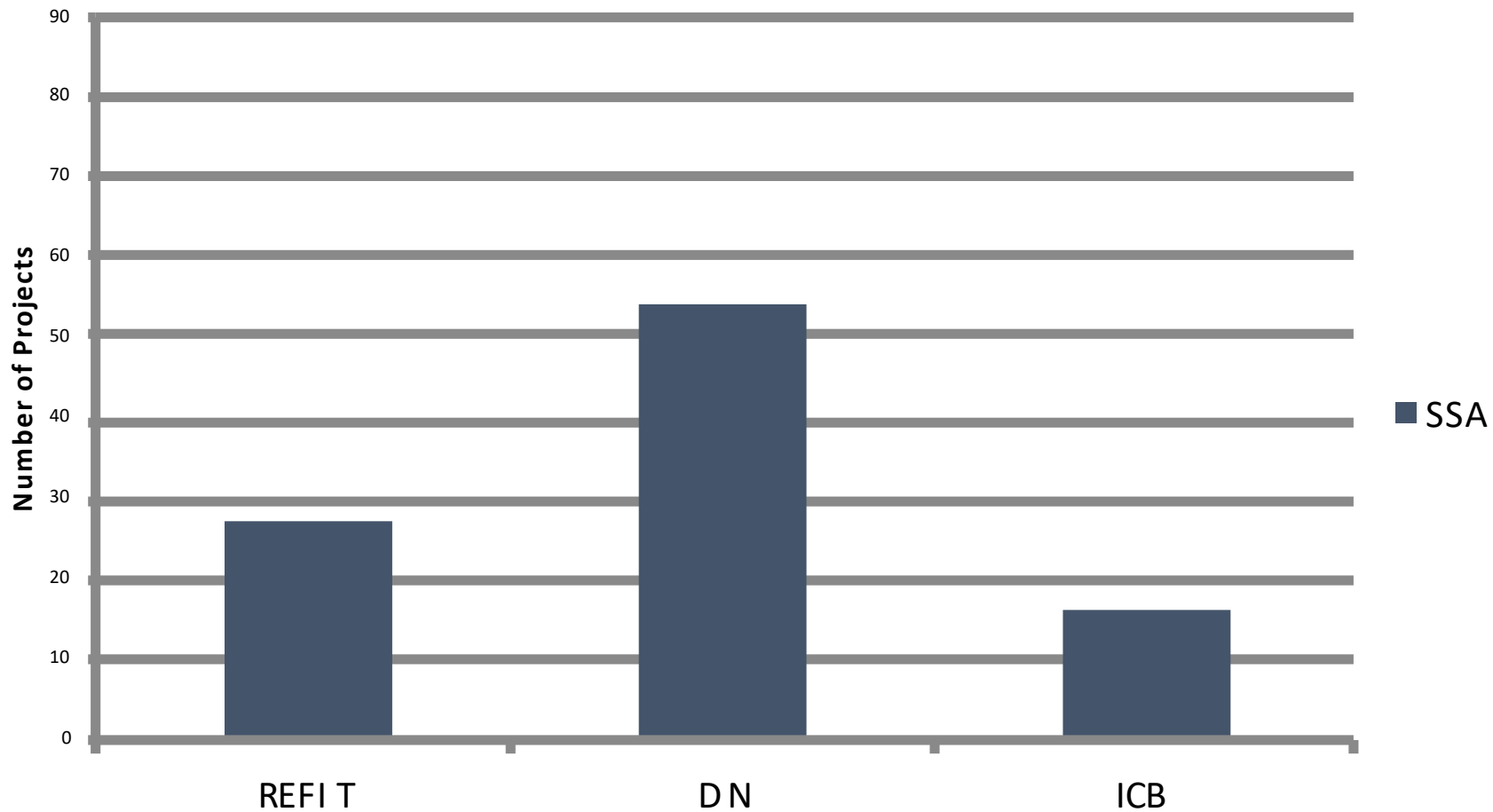
Lowest prices being achieved in emerging markets



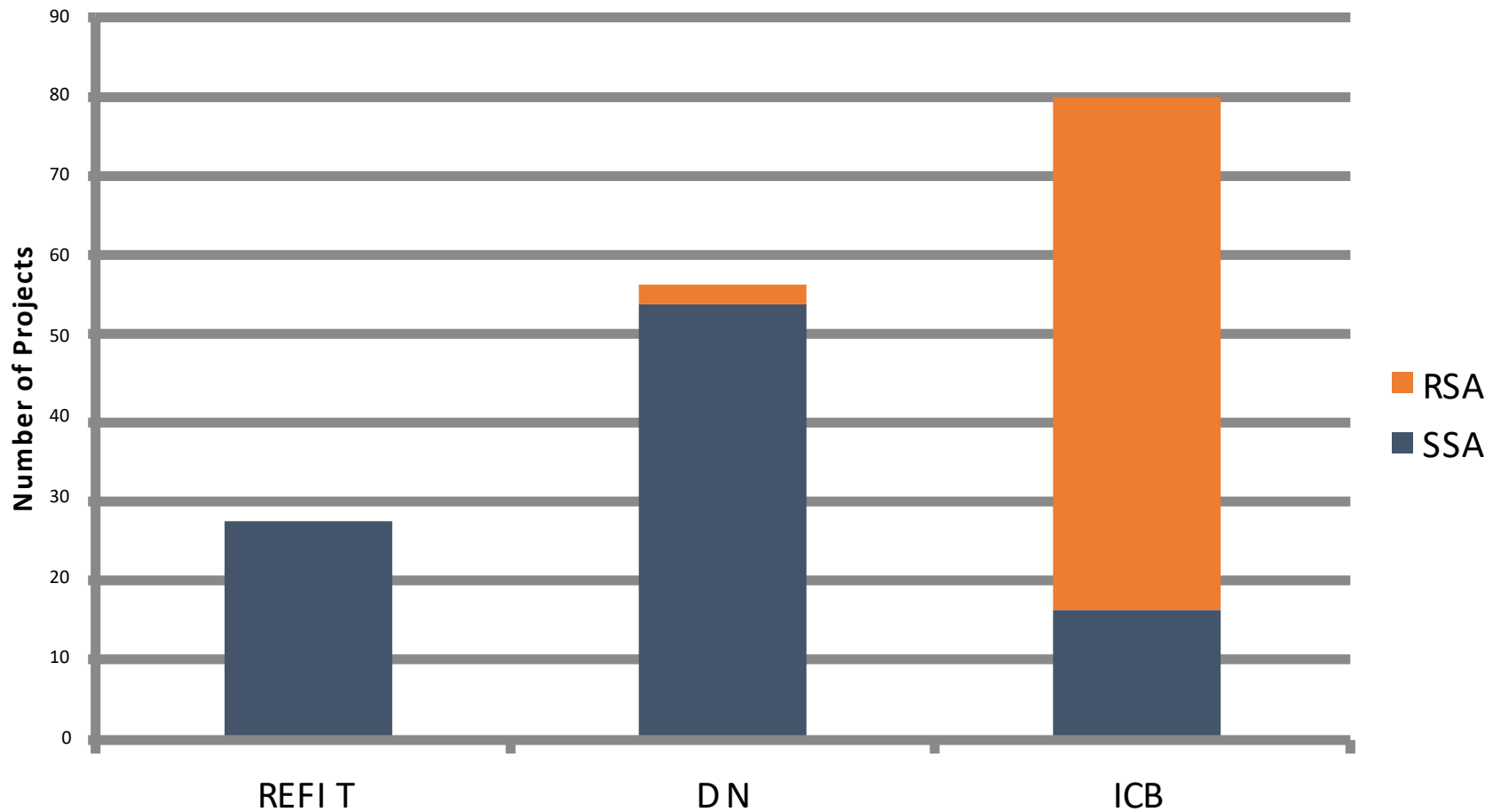
Is Africa benefitting?



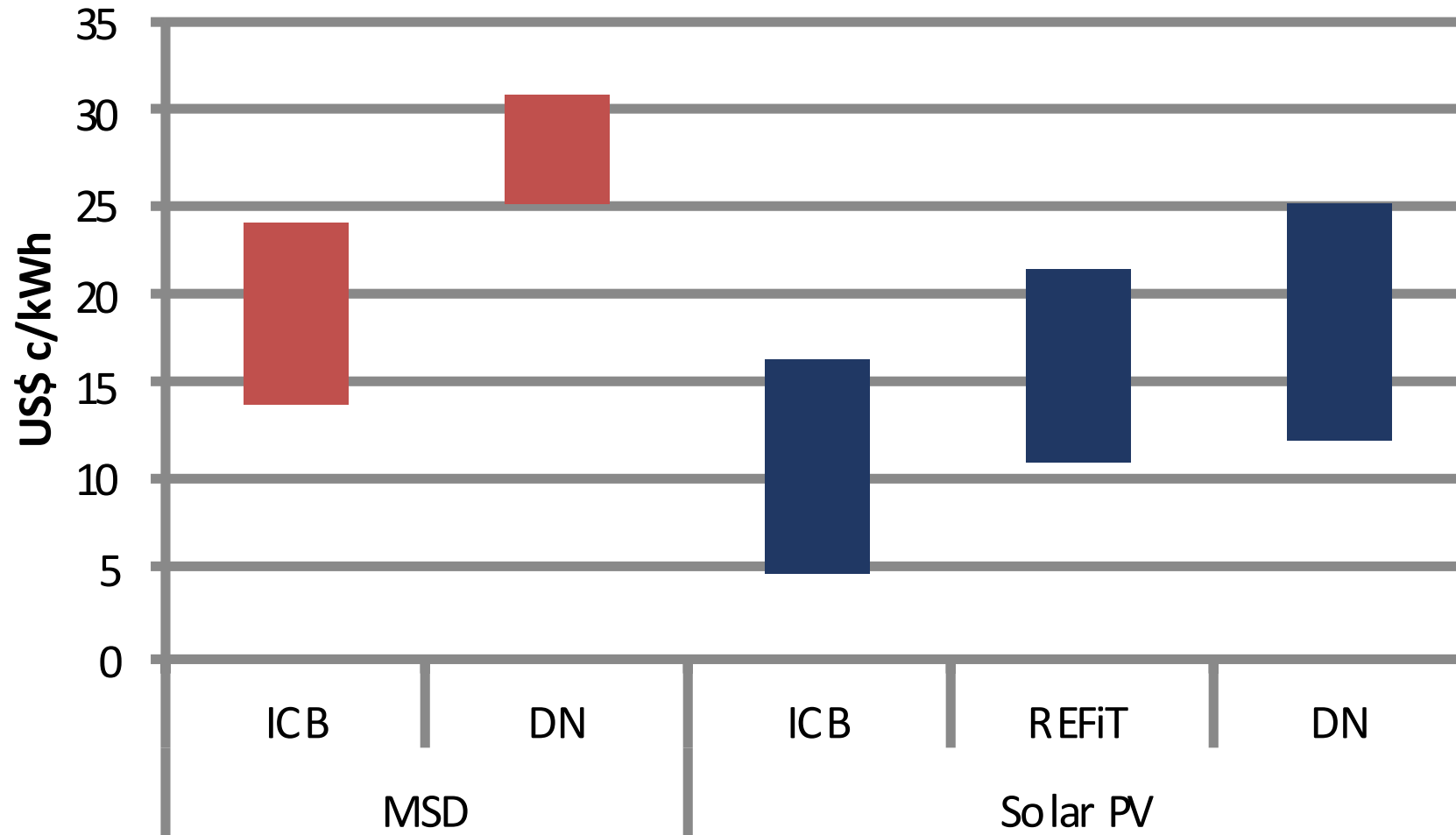
How are most IPPs procured in sub-Saharan Africa?



How are most IPPs procured in sub-Saharan Africa?



Which procurement methods yield better prices in sub-Saharan Africa?



Key issues to consider when designing auctions

What is being auctioned?

- Dynamic, least-cost plan ---> competitive procurement
- Volume
 - Fixed or price-sensitive?
- Technology neutral or specific technologies
- Energy (MWh) or capacity (MW)
 - Timeblocks?
- Site Selection & Preparation
 - Government or developers?

Who can take part & who wins?

- Qualification & compliance requirements
 - Reputation, capacity, **commitment**
 - Secured site, grid access, permits
- Winner selection criteria
 - Price only? Price + ?
- Price caps
 - Disclosed or undisclosed
- Project size/award caps
- Local economic development

Key issues to consider when designing auctions

How are investor/lender risks mitigated?

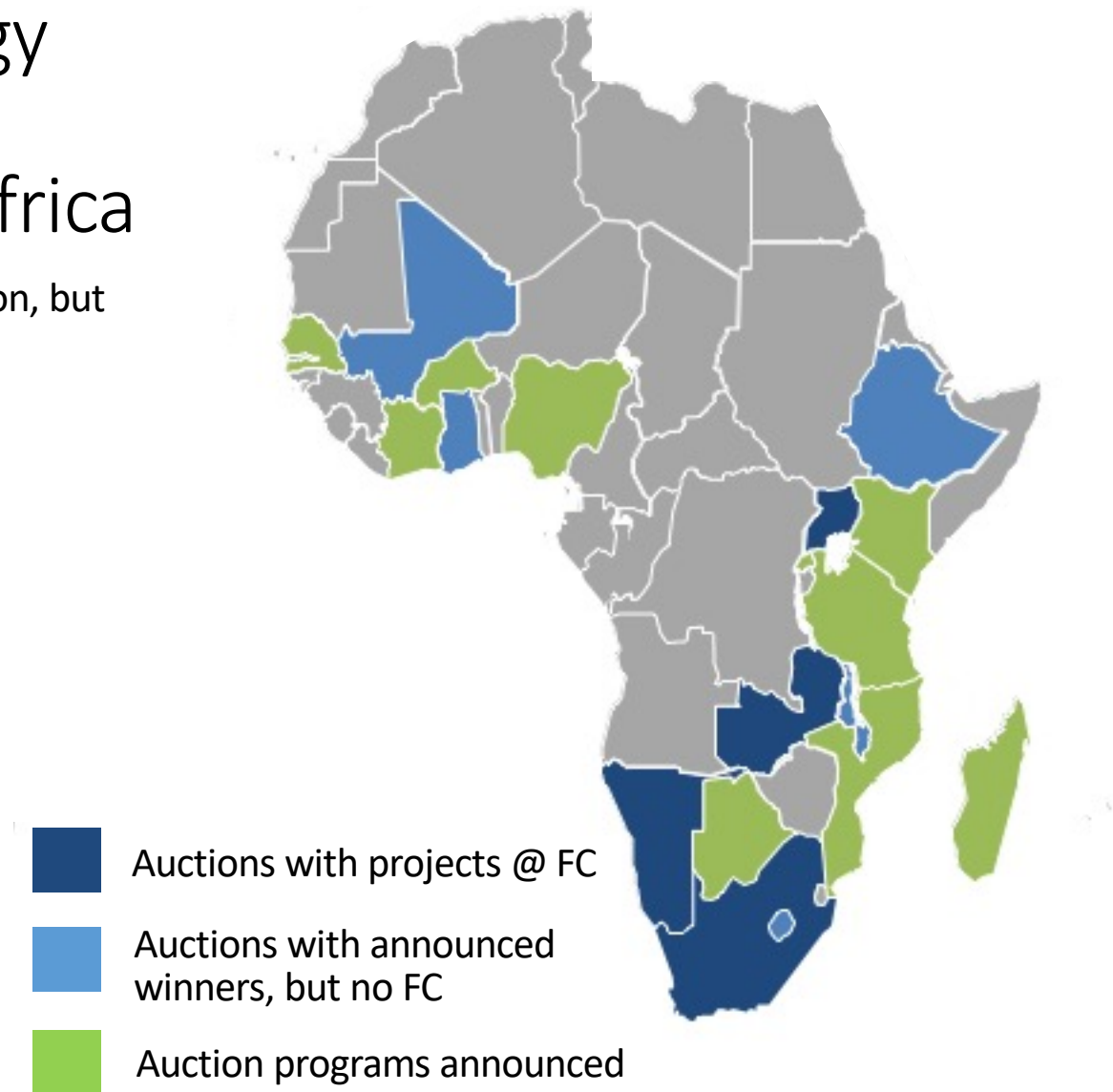
- Bankable documentation & contracts
 - Standardised, non-negotiable, tested
- Termination & liquidity guarantees, DFI involvement, arbitration arrangements
- PPA currency & indexation
 - Hard/local currency, full/partial/no indexation

How is the auction being implemented?

- Mandated, authorized & coordinated leadership
- Well-resourced procurement programme
 - Transaction advisors, ongoing costs
- Transmission planning
- Process
 - Transparent, thorough review process with external teams, audits
 - Stick to timelines
 - Communicate effectively

Renewable energy auction progress in sub-Saharan Africa

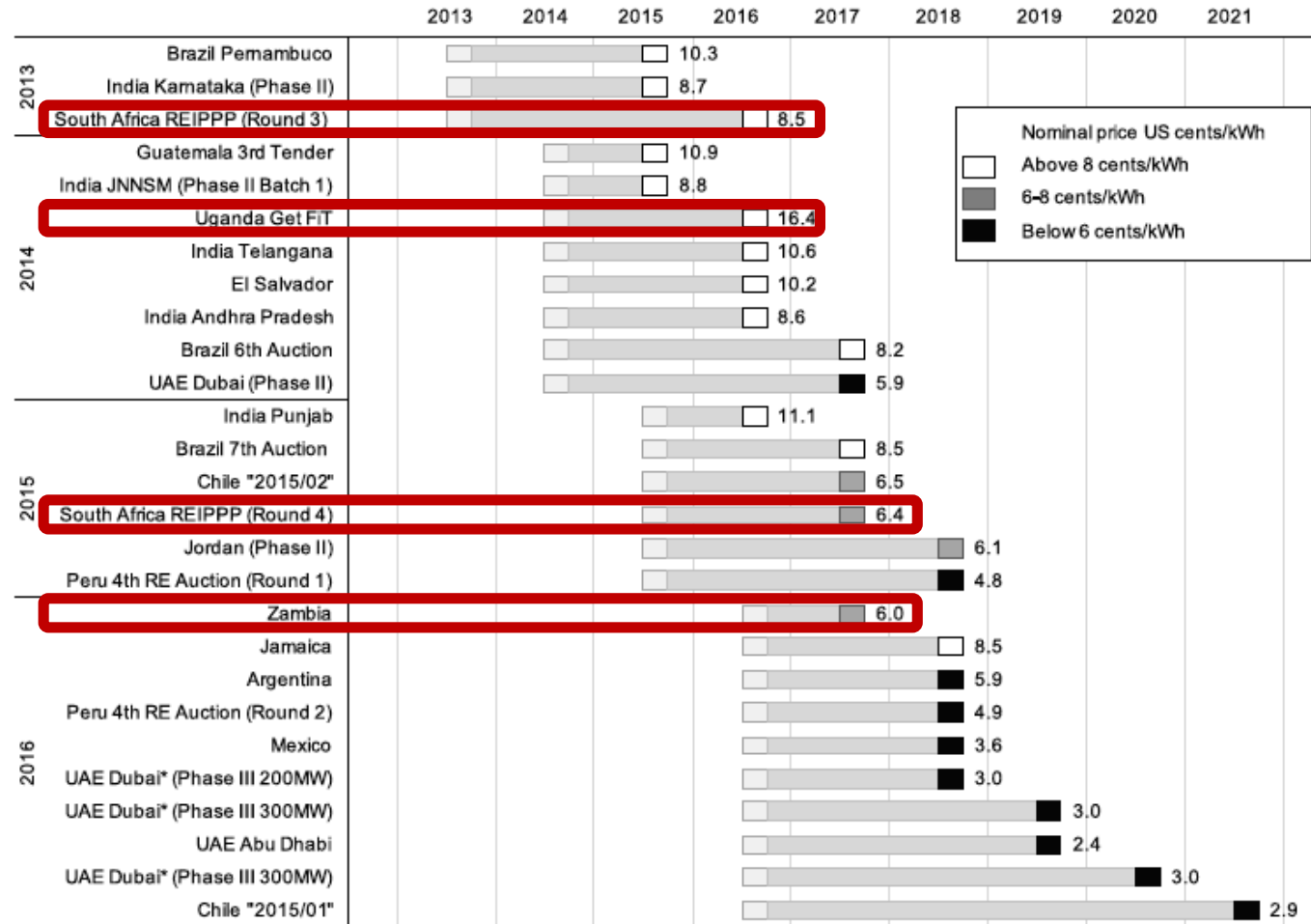
Rapidly spreading across the region, but
varying results



Features of renewable energy auctions in sub-Saharan Africa

	South Africa*	Uganda*	Zambia*	Ghana	Namibia*	Malawi	Ethiopia	Senegal
Year	2011 - 2018	2014	2016	2016	2017	2017	2017	2018
Auction Demand	6,300 MW (4 rounds) Multiple RE	4 x 5 MW Solar PV	2 x 50MW Solar PV	1 x 20MW Solar PV	1 x 37 MW Solar PV	Max 80 MW Solar PV (4x sites)	1 x 100 MW Solar PV	2 x 30 MW Solar PV
Site Selection	Developer	Developer (3km - grid)	Selected by govt.	Developer (input from offtaker)	Selected by govt./ utility	Substations identified by govt.	Selected by govt.	Selected by govt.
Local Content	40% min	None	None	20%	None (but 30% local share- holding)	5% develop ment & construction 20% O&M	15%	None
Evaluation	70:30 Price:Economic Development	70:30 Price:Technical	Price	Not clear	70: 30 Price:Technical	Price	70:30 Price:Technical	Price
PPA	20 Years	20 Years	25 Years	20 Years	20 Years	25 Years	20 Years	20 Years
Guarantees	Sovereign	Sovereign & Liquidity	Sovereign & Liquidity	Sovereign & Liquidity	None	Sovereign & Liquidity	Sovereign	Sovereign & Liquidity
Winning Price (USDc/ kWh)	4,7*	16,37	6,02	11,47	6,02	7,35 – 10,35 (TBC)	Below US\$c6 (TBC)	4,7
Currency	ZAR	US\$/EUR	US\$	US\$	NA\$	US\$	US\$	US\$
Financial Close	Yes	Yes	Yes*	No	Yes	No	No	No

A note on prices



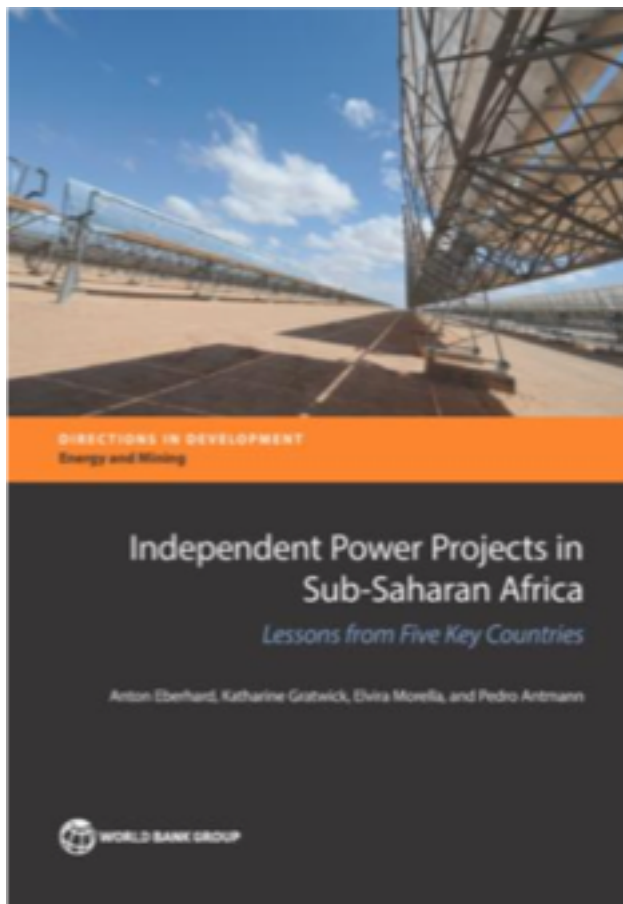
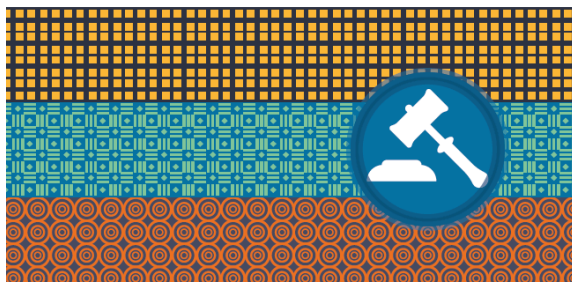
THE SOUTH AFRICAN RENEWABLE ENERGY IPP PROCUREMENT PROGRAMME

Review, Lessons Learned & Proposals to Reduce Transaction Costs

Anton Eberhard and Raine Naude



RENEWABLE ENERGY AUCTIONS — CASES FROM SUB-SAHARAN AFRICA



Wikus Kruger

Graduate School of Business

University of Cape Town

Wikus.kruger@gsb.uct.ac.za

The Management Programme in Infrastructure Reform & Regulation (MIR) is an emerging centre of excellence and expertise in Africa.

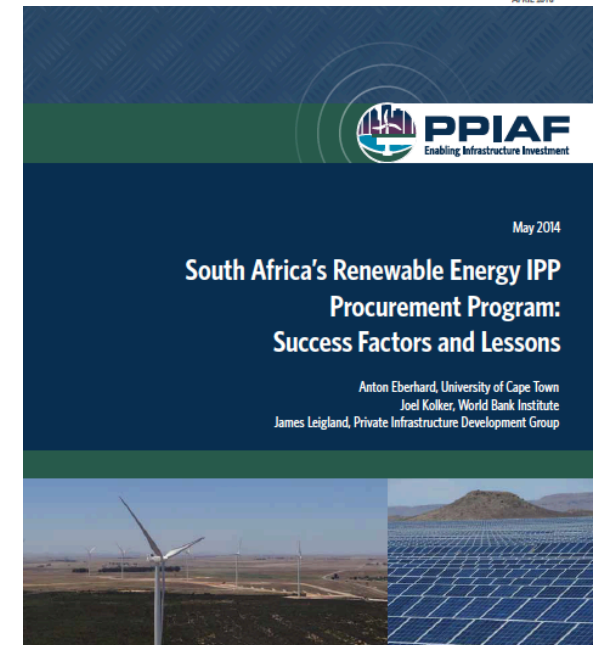
RECOMMENDATIONS FOR THE DESIGN OF SUCCESSFUL RENEWABLE ENERGY AUCTIONS OR COMPETITIVE TENDERS IN AFRICA

LESSONS FROM SOUTH AFRICA

Anton Eberhard and Raine Naude



APRIL 2016



May 2014

South Africa's Renewable Energy IPP Procurement Program: Success Factors and Lessons

Anton Eberhard, University of Cape Town
Joel Kolker, World Bank Institute
James Leigland, Private Infrastructure Development Group

